

New concepts
BP measurement Wider use of out-of-office BP measurement with ABPM and/or HBPM , especially HBPM, as an option to confirm the diagnosis of hypertension, detect white-coat and masked hypertension and monitor BP control
Less conservative treatment of BP in older and very old patients Lower BP thresholds and treatment targets for older patients , with emphasis on considerations of biological rather than chronological age (i.e. the importance of frailty, independence, and the tolerability of treatment) Recommendation that treatment should never be denied or withdrawn on the basis of age , provided that treatment is tolerated
A SPC treatment strategy to improve BP control Preferred use of two-drug combination therapy for the initial treatment of most people with hypertension A single-pill treatment strategy for hypertension with the preferred use of SPC therapy for most patients Simplified drug treatment algorithms with the preferred use of an ACE inhibitor or ARB, combined with a CCB and/or a thiazide/thiazide-like diuretic, as the core treatment strategy for most patients, with beta-blockers used for specific indications
New target ranges for BP in treated patients Target BP ranges for treated patients to better identify the recommended BP target and lower safety boundaries for treated BP , according to a patient's age and specific comorbidities
Detecting poor adherence to drug therapy A strong emphasis on the importance of evaluating treatment adherence as a major cause of poor BP control
A key role for nurses and pharmacists in the longer-term management of hypertension The important role of nurses and pharmacists in the education, support, and follow-up of treated hypertensive patients is emphasized as part of the overall strategy to improve BP control

ABPM, ambulatory blood pressure monitoring; ACE, angiotensin-converting enzyme; AF, atrial fibrillation; ARB, angiotensin receptor blocker; BP, blood pressure; CCB, calcium channel blocker; CVD, cardiovascular disease; HBPM, home blood pressure monitoring; HMOD, hypertension-mediated organ damage; SCORE, Systematic COronary Risk Evaluation; SPC, single-pill combination.

3 DEFINITION, CLASSIFICATION, AND EPIDEMIOLOGICAL ASPECTS OF HYPERTENSION

3.1 Definition of hypertension

The relationship between BP and cardiovascular and renal events is continuous, making the distinction between normotension and hypertension, based on cut-off BP values, somewhat arbitrary [2,4,8]. However, in practice, cut-off BP values are used for pragmatic reasons to simplify the diagnosis and decisions about treatment. Epidemiological associations between BP and cardiovascular risk extend from very low levels of BP (i.e. SBP >115 mmHg). However, ‘hypertension’ is defined as the level of BP at which the benefits of treatment (either with lifestyle interventions or drugs) unequivocally outweigh the risks of treatment, as documented by clinical trials. This evidence has been reviewed (see Section 7.2 for detailed discussion of hypertension diagnostic thresholds) and provides the basis for

TABLE 3. Classification of office blood pressure^a and definitions of hypertension grade^b

Category	Systolic (mmHg)		Diastolic (mmHg)
Optimal	<120	and	<80
Normal	120–129	and/or	80–84
High normal	130–139	and/or	85–89
Grade 1 hypertension	140–159	and/or	90–99
Grade 2 hypertension	160–179	and/or	100–109
Grade 3 hypertension	≥180	and/or	≥110
Isolated systolic hypertension ^b	≥140	and	<90

BP, blood pressure.
^aBP category is defined according to seated clinic BP and by the highest level of BP, whether systolic or diastolic.
^bIsolated systolic hypertension is graded 1, 2, or 3 according to systolic BP values in the ranges indicated. The same classification is used for all ages from 16 years.

the recommendation that the classification of BP and definition of hypertension remain unchanged from previous ESH/ESC Guidelines (Table 3) [15–17].

Hypertension is defined as office SBP values at least 140 mmHg and/or diastolic BP (DBP) values at least 90 mmHg. This is based on evidence from multiple RCTs that treatment of patients with these BP values is beneficial (see Section 7). The same classification is used in younger, middle-aged, and older people, whereas BP centiles are used in children and teenagers, in whom data from interventional trials are not available. Details on BP classification in boys and girls 16 years or less of age can be found in the 2016 ESH Guidelines for children and adolescents [18].

3.2 Classification of blood pressure

Classification of BP

Recommendation	Class ^a	Level ^b
It is recommended that BP be classified as optimal, normal, high-normal, or grades 1–3 hypertension, according to office BP.	I	C

BP, blood pressure.
^aClass of recommendation.
^bLevel of evidence.

3.3 Prevalence of hypertension

Based on office BP, the global prevalence of hypertension was estimated to be 1.13 billion in 2015 [5], with a prevalence of over 150 million in central and Eastern Europe. The overall prevalence of hypertension in adults is around 30–45% [12], with a global age-standardized prevalence of 24 and 20% in men and women, respectively, in 2015 [5]. This high prevalence of hypertension is consistent across

the world, irrespective of income status, that is in lower, middle and higher income countries [12]. Hypertension becomes progressively more common with advancing age, with a prevalence of more than 60% in people aged more than 60 years [12]. As populations age, adopt more sedentary lifestyles, and increase their body weight, the prevalence of hypertension worldwide will continue to rise. It is estimated that the number of people with hypertension will increase by 15–20% by 2025, reaching close to 1.5 billion [19].

3.4 Blood pressure relationship with risk of cardiovascular and renal events

Elevated BP was the leading global contributor to premature death in 2015, accounting for almost 10 million deaths and over 200 million disability-adjusted life years [3]. Importantly, despite advances in diagnosis and treatment over the past 30 years, the disability-adjusted life years attributable to hypertension have increased by 40% since 1990 [3]. SBP at least 140 mmHg accounts for most of the mortality and disability burden (70%), and the largest number of SBP-related deaths per year are due to ischaemic heart disease (4.9 million), haemorrhagic stroke (2.0 million), and ischaemic stroke (1.5 million) [3].

Both office BP and out-of-office BP have an independent and continuous relationship with the incidence of several cardiovascular events [haemorrhagic stroke, ischaemic stroke, myocardial infarction, sudden death, heart failure, and peripheral artery disease (PAD)], as well as end-stage renal disease [4]. Accumulating evidence is closely linking hypertension with an increased risk of developing atrial fibrillation (AF) [20], and evidence is emerging that links early elevations of BP to increased risk of cognitive decline and dementia [21,22].

The continuous relationship between BP and risk of events has been shown at all ages [23] and in all ethnic groups [24,25], and extends from high BP levels to relatively low values. SBP appears to be a better predictor of events than DBP after the age of 50 years [23,26,27]. High DBP is associated with increased cardiovascular risk and is more commonly elevated in younger (<50 years) vs. older patients. DBP tends to decline from midlife as a consequence of arterial stiffening; consequently, SBP assumes even greater importance as a risk factor from midlife [26]. In middle-aged and older people, increased pulse pressure (the difference between SBP and DBP values) has additional adverse prognostic significance [28,29].

3.5 Hypertension and total cardiovascular risk assessment

Hypertension rarely occurs in isolation, and often clusters with other cardiovascular risk factors such as dyslipidaemia and glucose intolerance [30,31]. This metabolic risk factor clustering has a multiplicative effect on cardiovascular risk [32]. Consequently, quantification of total cardiovascular risk (i.e. the likelihood of a person developing a cardiovascular event over a defined period) is an important part of the risk stratification process for patients with hypertension.

Many cardiovascular risk assessment systems are available and most project 10-year risk. Since 2003, the

European Guidelines on CVD prevention have recommended use of the Systematic Coronary Risk Evaluation (SCORE) system because it is based on large, representative European cohort data sets (available at: <http://www.escardio.org/-Education/Practice-tools/CVD-prevention-toolbox/SCORE-Risk-Charts>). The SCORE system estimates the 10-year risk of a first fatal atherosclerotic event, in relation to age, sex, smoking habits, total cholesterol level, and SBP. The SCORE system also allows calibration for different cardiovascular risk levels across numerous European countries and has been externally validated [33]. A previous limitation of the SCORE system was that it applied only to patients aged 40–65 years; however, the SCORE system has recently been adapted for patients over the age of 65 years [34]. Detailed information on cardiovascular risk assessment is available [35].

Factors influencing cardiovascular risk factors in patients with hypertension are shown in Table 4. Hypertensive

TABLE 4. Factors influencing cardiovascular risk in patients with hypertension

Demographic characteristics and laboratory parameters
Sex ^a (men >women)
Age ^a
Smoking (current or past history) ^a
Total cholesterol ^a and HDL-C
Uric acid
Diabetes ^a
Overweight or obesity
Family history of premature CVD (men aged <55 years and women aged <65 years)
Family or parental history of early-onset hypertension
Early-onset menopause
Sedentary lifestyle
Psychosocial and socioeconomic factors
Heart rate (resting values >80 beats/min)
Asymptomatic HMOD
Arterial stiffening: Pulse pressure (in older people) ≥60 mmHg Carotid–femoral PWV >10 m/s
ECG LVH (Sokolow–Lyon index >35 mm, or R in aVL ≥11 mm; Cornell voltage duration product >2440 mm ² ·ms, or Cornell voltage >28 mm in men or >20 mm in women)
Echocardiographic LVH [left ventricular mass index: men >50 g/m ^{2.7} ; women >47 g/m ^{2.7} (height in m ^{2.7}); indexation for BSA may be used in normal-weight patients; left ventricular mass/BSA g/m ² >115 (men) and >95 (women)]
Microalbuminuria (30–300 mg/24 h), or elevated albumin–creatinine ratio (30–300 mg/g; 3.4–34 mg/mmol) (preferentially on morning spot urine) ^b
Moderate CKD with eGFR 30–59 ml/min/1.73 m ² (BSA) ^b
Ankle-brachial index <0.9
Advanced retinopathy: haemorrhages or exudates, papilloedema
Established cardiovascular or renal disease
Cerebrovascular disease: ischaemic stroke, cerebral haemorrhage, TIA
CAD: myocardial infarction, angina, myocardial revascularization
Presence of atheromatous plaque on imaging
Heart failure, including HFpEF
Peripheral artery disease
Atrial fibrillation
Severe CKD with eGFR <30 ml/min/1.73 m ²

BSA, body surface area; CAD, coronary artery disease; CKD, chronic kidney disease; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; HDL-C, HDL cholesterol; HFpEF, heart failure with preserved ejection fraction; HMOD, hypertension-mediated organ damage; LVH, left ventricular hypertrophy; PWV, pulse wave velocity; SCORE, Systematic COronary Risk Evaluation; TIA, transient ischaemic attack.

^aCV risk factors included in the SCORE system.

^bProteinuria and reduced eGFR are independent risk factors. See Table 6 for cardiovascular risk modifiers.

patients with documented CVD, including asymptomatic atheromatous disease on imaging, type 1 or type 2 diabetes, very high levels of individual risk factors (including grade 3 hypertension), or chronic kidney disease (CKD; stages 3–5), are automatically considered to be at very high (i.e. $\geq 10\%$ CVD mortality) or high (i.e. 5–10% CVD mortality) 10 year cardiovascular risk (Table 5). Such patients do not need formal cardiovascular risk estimation to determine their need for treatment of their hypertension and other cardiovascular risk factors. For all other hypertensive patients, estimation of 10-year cardiovascular risk using the SCORE system is recommended. Estimation should be complemented by assessment of hypertension-mediated organ damage (HMOD), which can also increase cardiovascular risk to a higher level, even when asymptomatic (see Table 4 and Sections 3.6 and 4).

There is also emerging evidence that an increase in serum uric acid to levels lower than those typically associated with gout is independently associated with increased cardiovascular risk in both the general population and in hypertensive patients. Measurement of serum uric acid is recommended as part of the screening of hypertensive patients [36].

The SCORE system only estimates the risk of fatal cardiovascular events. The risk of total cardiovascular events (fatal and nonfatal) is approximately three times higher

than the rate of fatal cardiovascular events in men and four times higher in women. This multiplier is attenuated to less than three times in older people in whom a first event is more likely to be fatal [37].

There are important general modifiers of cardiovascular risk (Table 6) as well as specific cardiovascular risk modifiers for patients with hypertension. Cardiovascular risk modifiers are particularly important at the cardiovascular risk boundaries, and especially for patients at moderate-risk in whom a risk modifier might convert moderate-risk to high risk and influence treatment decisions with regard to cardiovascular risk factor management. Furthermore, cardiovascular risk estimates by the SCORE system may be modified in first-generation immigrants to Europe and cardiovascular risk scores in such patients may be adjusted by correction factors (Table 7). Further details of the impact of cardiovascular risk modifiers are available from the ESC 2016 CVD prevention Guidelines [35].

3.6 Importance of hypertension-mediated organ damage in refining cardiovascular risk assessment in hypertensive patients

A unique and important aspect of cardiovascular risk estimation in hypertensive patients is the need to consider

TABLE 5. Ten year cardiovascular risk categories (Systematic COronary Risk Evaluation system)

Very high risk	People with any of the following:
	Documented CVD, either clinical or unequivocal on imaging . • Clinical CVD includes acute myocardial infarction, acute coronary syndrome, coronary or other arterial revascularization, stroke, TIA, aortic aneurysm, and PAD • Unequivocal documented CVD on imaging includes significant plaque (i.e. $\geq 50\%$ stenosis) on angiography or ultrasound; it does not include increase in carotid intima-media thickness • Diabetes mellitus with target organ damage, e.g. proteinuria or a with a major risk factor such as grade 3 hypertension or hypercholesterolaemia • Severe CKD (eGFR < 30 mL/min/1.73 m²) • A calculated 10 year SCORE of $\geq 10\%$
High risk	People with any of the following:
	• Marked elevation of a single risk factor, particularly cholesterol > 8 mmol/L (> 310 mg/dL), e.g. familial hypercholesterolaemia or grade 3 hypertension (BP $\geq 180/110$ mmHg) • Most other people with diabetes mellitus (except some young people with type 1 diabetes mellitus and without major risk factors, who may be at moderate-risk)
	Hypertensive LVH
	Moderate CKD (eGFR 30–59 mL/min/1.73 m²)
Moderate risk	A calculated 10 year SCORE of 5–10 %
	People with: • A calculated 10 year SCORE of ≥ 1 to $< 5\%$ • Grade 2 hypertension • Many middle-aged people belong to this category
Low risk	People with: • A calculated 10 year SCORE of $< 1\%$

BP, blood pressure; CKD, chronic kidney disease; CVD, cardiovascular disease; eGFR, estimated glomerular filtration rate; LVH, left ventricular hypertrophy; TIA, transient ischaemic attack; PAD, peripheral artery disease; SCORE, Systematic COronary Risk Evaluation.

TABLE 6. Risk modifiers increasing cardiovascular risk estimated by the Systemic COronary Risk Evaluation (SCORE) system [35]

Social deprivation, the origin of many causes of CVD
Obesity (measured by BMI) and central obesity (measured by waist circumference)
Physical inactivity
Psychosocial stress, including vital exhaustion
Family history of premature CVD (occurring at age <55 years in men and <60 years in women)
Autoimmune and other inflammatory disorders
Major psychiatric disorders
Treatment for infection with human immunodeficiency virus
Atrial fibrillation
Left ventricular hypertrophy
CKD
Obstructive sleep apnoea syndrome

BMI, body mass index; CKD, chronic kidney disease; CVD, cardiovascular disease.

the impact of HMOD. This was previously termed ‘target organ damage’, but HMOD more accurately describes hypertension-induced structural and/or functional changes in major organs (i.e. the heart, brain, retina, kidney, and vasculature) (Table 4). There are three important considerations: not all features of HMOD are included in the SCORE system (CKD and established vascular disease are included) and several hypertensive HMODs (e.g. cardiac, vascular, and retinal) have well established adverse prognostic significance (see Section 5) and may, especially if HMOD is pronounced, lead to a high cardiovascular risk even in the absence of classical cardiovascular risk factors; the presence of HMOD is common and often goes undetected [38]; and the presence of multiple HMODs in the same patient is also common, and further increases cardiovascular risk [39–41]. Consequently, the inclusion of HMOD assessment is important in patients with hypertension and helps identify high-risk or very high-risk hypertensive patients who may otherwise be misclassified as having a lower level of risk by the SCORE system [42]. This is especially true for the presence of left ventricular hypertrophy (LVH), CKD with albuminuria or proteinuria, or arterial stiffening [43] (see Section 5). The impact of progression of the stages of hypertension-associated disease (from uncomplicated through to asymptomatic or established disease), according to different grades of hypertension and the presence of cardiovascular risk factors, HMOD, or comorbidities, is illustrated in Figure 1 for middle-aged individuals.

TABLE 7. Correction factors for the Systemic COronary Risk Evaluation (SCORE) cardiovascular risk estimates in first-generation immigrants to Europe [35]

Region of origin	Multiplication factor
Southern Asia	1.4
Sub-Saharan Africa	1.3
Caribbean	1.3
Western Asia	1.2
Northern Africa	0.9
Eastern Asia	0.7
Southern America	0.7

3.7 Challenges in cardiovascular risk assessment

Cardiovascular risk is strongly influenced by age (i.e. older people are invariably at high absolute cardiovascular risk). In contrast, the absolute risk of younger people, particularly younger women, is invariably low, even in those with a markedly abnormal risk factor profile. In the latter, relative risk is elevated even if absolute risk is low. The use of ‘cardiovascular risk age’ has been proposed as a useful way of communicating risk and making treatment decisions, especially for younger people at low absolute risk but with high relative risk [35]. This works by illustrating how a younger patient (e.g. a 40-year-old) with risk factors but low absolute risk has a cardiovascular risk equivalent to a much older person (60 years) with optimal risk factors; thus, the cardiovascular risk age of the younger patient is 60 years. The cardiovascular risk age can be automatically calculated using HeartScore (www.heartscore.org).

A second consideration is that the presence of concomitant disease is often recorded in a binary way in cardiovascular risk assessment systems (e.g. diabetes, yes/no). This does not reflect the impact of the severity or duration of concomitant diseases on total cardiovascular risk. For example, long-standing diabetes is clearly associated with high risk, whereas the risk is less certain for recent-onset diabetes [34].

A third conundrum specific to hypertension is what BP value to use in cardiovascular risk assessment in a patient who is receiving treatment for hypertension. If treatment was commenced recently, it seems appropriate to use the pretreatment BP value. If treatment has been long-standing, using the current treated BP value will invariably underestimate risk because it does not reflect prior longer-term exposure to higher BP levels, and antihypertensive treatment does not completely reverse the risk even when BP is well controlled. If treatment has been long-standing, then the ‘treated BP value’ should be used, with the caveat that the calculated cardiovascular risk will be lower than the patient’s actual risk. A fourth conundrum is how to impute out-of-office BP values into risk calculators that have been calibrated according to office BP readings. These various limitations should be kept in mind when estimating cardiovascular risk in clinical practice.

Hypertension and cardiovascular risk assessment

Recommendation	Class ^a	Level ^b
CV risk assessment with the SCORE system is recommended for hypertensive patients who are not already at high or very high risk due to established CVD, renal disease, or diabetes, a markedly elevated single risk factor (e.g. cholesterol), or hypertensive LVH [33,35].	I	B

CVD, cardiovascular disease; LVH, left ventricular hypertrophy; SCORE, Systemic COronary Risk Evaluation.

^aClass of recommendation.

^bLevel of evidence.

Hypertension disease staging	Other risk factors, HMOD, or disease	BP (mmHg) grading			
		High normal SBP 130–139 DBP 85–89	Grade 1 SBP 140–159 DBP 90–99	Grade 2 SBP 160–179 DBP 100–109	Grade 3 SBP ≥180 or DBP ≥110
Stage 1 (uncomplicated)	No other risk factors	Low risk	Low risk	Moderate risk	High risk
	1 or 2 risk factors	Low risk	Moderate risk	Moderate to high risk	High risk
	≥ 3 risk factors	Low to Moderate risk	Moderate to high risk	High Risk	High risk
Stage 2 (asymptomatic disease)	HMOD, CKD grade 3, or diabetes mellitus without organ damage	Moderate to high risk	High risk	High risk	High to very high risk
Stage 3 (established disease)	Established CVD, CKD grade ≥ 4, or diabetes mellitus with organ damage	Very high risk	Very high risk	Very high risk	Very high risk

FIGURE 1 Classification of hypertension stages according to blood pressure levels, presence of cardiovascular risk factors, hypertension-mediated organ damage, or comorbidities. Cardiovascular risk is illustrated for a middle-aged male. The cardiovascular risk does not necessarily correspond to the actual risk at different ages. The use of the SCORE system is recommended for formal estimation of cardiovascular risk for treatment decisions. BP, blood pressure; CKD, chronic kidney disease; DBP, diastolic blood pressure; HMOD, hypertension-mediated organ damage; SBP, systolic blood pressure; SCORE, Systematic COronary Risk Evaluation.

4 BLOOD PRESSURE MEASUREMENT

4.1 Conventional office blood pressure measurement

Auscultatory or oscillometric semiautomatic or automatic sphygmomanometers are the preferred method for measuring BP in the doctor’s office. These devices should be validated according to standardized conditions and protocols [44]. BP should initially be measured in both upper arms, using an appropriate cuff size for the arm circumference. A consistent and significant SBP difference between arms (i.e. >15 mmHg) is associated with an increased cardiovascular risk [45], most likely due to atheromatous vascular disease. Where there is a difference in BP between arms, ideally established by simultaneous measurement, the arm with the higher BP values should be used for all subsequent measurements.

In older people, people with diabetes, or people with other causes of orthostatic hypotension, BP should also be measured 1 and 3 min after standing. Orthostatic hypotension is defined as a reduction in SBP of at least 20 mmHg or in DBP of at least 10 mmHg within 3 min of standing, and is associated with an increased risk of mortality and cardiovascular events [46]. Heart rate should also be recorded at the time of BP measurements because resting heart rate is an independent predictor of cardiovascular morbid or fatal events [47], although heart rate is not included in any cardiovascular risk algorithm. Table 8 summarizes the recommended procedure for routine office BP measurement. It is emphasized that office BP is often performed improperly, with inadequate attention to the standardized conditions recommended for a valid measurement of office BP. Improper measurement of office BP can lead to inaccurate

classification, overestimation of a patient’s true BP, and unnecessary treatment.

4.2 Unattended office blood pressure measurement

Automated multiple BP readings in the doctor’s office improve the reproducibility of BP measurement, and if the patient is seated alone and unobserved, the ‘white-coat

TABLE 8. Office blood pressure measurement

Patients should be seated comfortably in a quiet environment for 5 min before beginning BP measurements.
Three BP measurements should be recorded, 1–2 min apart, and additional measurements only if the first two readings differ by >10 mmHg. BP is recorded as the average of the last two BP readings.
Additional measurements may have to be performed in patients with unstable BP values due to arrhythmias, such as in patients with AF, in whom manual auscultatory methods should be used as most automated devices have not been validated for BP measurement in patients with AF. ^a
Use a standard bladder cuff (12–13 cm wide and 35 cm long) for most patients, but have larger and smaller cuffs available for larger (arm circumference >32 cm) and thinner arms, respectively.
The cuff should be positioned at the level of the heart, with the back and arm supported to avoid muscle contraction and isometric exercise-dependent increases in BP.
When using auscultatory methods, use phase I and V (sudden reduction/disappearance) Korotkoff sounds to identify SBP and DBP, respectively.
Measure BP in both arms at the first visit to detect possible between-arm differences. Use the arm with the higher value as the reference.
Measure BP 1 min and 3 min after standing from a seated position in all patients at the first measurement to exclude orthostatic hypotension. Lying and standing BP measurements should also be considered in subsequent visits in older people, people with diabetes, and people with other conditions in which orthostatic hypotension may frequently occur.
Record heart rate and use pulse palpation to exclude arrhythmia.

AF, atrial fibrillation; BP, blood pressure.
^aMost automatic devices are not validated for BP measurement in patients with AF and will record the highest individual systolic pressure wave form rather than an average of several cardiac cycles. This will lead to overestimation of BP.